GATE-AII BRANCHES
Engineering Mathematics



SINGLE VARIABLE CALCULUS

Lecture No.-10



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Recap of previous lecture



Topic Rolles theorem

Topic Lagrange mean value theorem

Topic Problems based on mean value theorem

Topic Basic integration

Topics to be covered



Topic Integration via parts

Topic Definite integrals

Topic Improper integrals

Topic Area bounded regions

Property (2) No.

If $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$ "DEFAULT Property" (KINGS)

$$\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$$

(3) $\int_{-a}^{a} f(x)\,dx = 0$ $f(x)$ is odd Function

$= 2\int_0^a f(x)\,dx$ $f(x)$ is even Function

odd function

$f(-x) = -f(x)$



Even Function

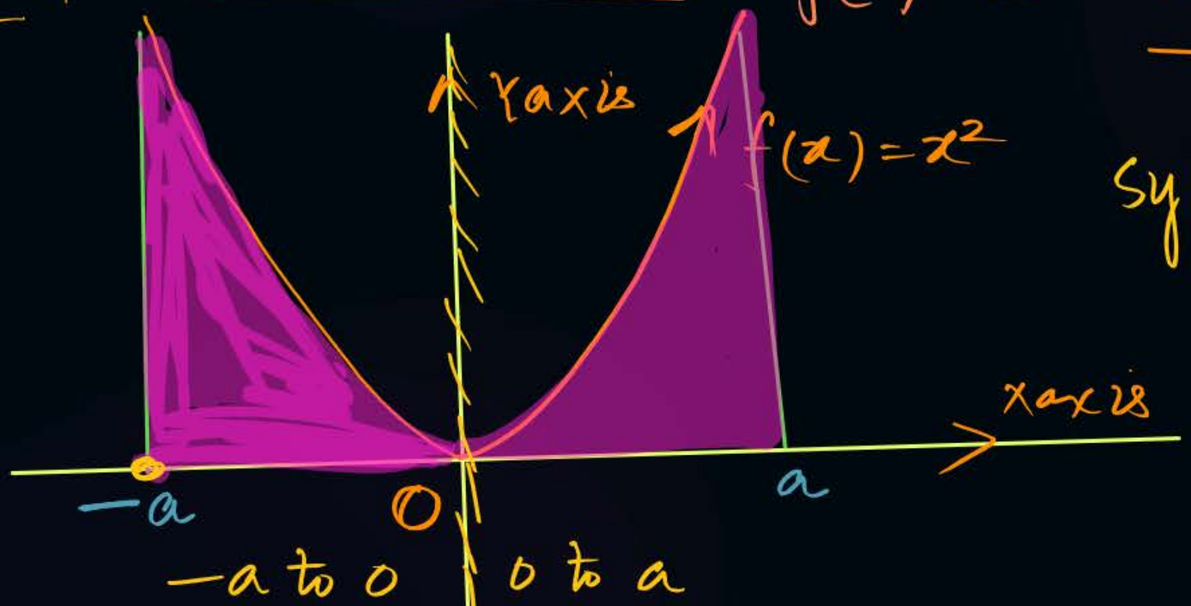
$f(-x) = f(x)$



Imp #

Even Function

$f(x) = x^2$



Area	+
AREA	−

Even Function $\xrightarrow{-x}$ Even x^2

Symmetric about Y axis

$f(x) = (-x)^2 = x^2$

$$\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$$

$= 2 \times$ Area of right side

odd function

$y = x^3 = f(x)$

$f(-x) = (-x)^3 = -x^3$

$$\int_{-a}^{a} f(x)\,dx = 0$$





Topic : Single Variable Calculus

#Q. The value of the integral $\int_0^{\pi/2} \frac{\sqrt{\cot x}}{\sqrt{\cot x}+\sqrt{\tan x}}\,dx$ is

Using Default Property

$$\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$$

$$I = \int_0^{\frac{\pi}{2}} \frac{\sqrt{\cot x}}{\sqrt{\cot x}+\sqrt{\tan x}}\,dx$$

$$+$$

$$I = \int_0^{\pi/2} \frac{\sqrt{\cot\left(\frac{\pi}{2}-x\right)}}{\sqrt{\cot\left(\frac{\pi}{2}-x\right)}+\sqrt{\tan\left(\frac{\pi}{2}-x\right)}} = \int_0^{\pi/2} \frac{\sqrt{\tan x}}{\sqrt{\tan x}+\sqrt{\cot x}}\,dx$$

$$2I = \int_0^{\pi/2} \frac{\sqrt{\tan x}+\sqrt{\cot x}}{\sqrt{\tan x}+\sqrt{\cot x}}\,dx = \int_0^{\pi/2} dx$$

$$2I = \frac{\pi}{2}$$

$$I = \pi/4$$

A. $\frac{\pi}{4}$

B. $\frac{\pi}{2}$

C. π

D. None of these

$$\int_0^a \frac{f(x)}{f(a-x)+f(x)}\,dx = \frac{a-0}{2}$$

$$\boxed{\int_a^b \frac{f(x)}{f(a+b-x)+f(x)}\,dx = \frac{(b-a)}{2}}$$

Trick

$$\int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}} = \frac{\frac{\pi}{2}-0}{2} = \frac{\pi}{4}$$

$$\int_5^{10} \frac{\sqrt{x}}{\sqrt{15-x}+\sqrt{x}}\,dx = = \frac{10-5}{2} = \frac{5}{2}$$

maximum time Ask gate ✓



Topic : Single Variable Calculus

#Q. Let $f(x) = \int_1^x \sqrt{2 - t^2}\, dt$ Then the real roots of the equation $x^2 - f'(x) = 0$ are

[2002 S]

A ± 1

B $\pm 1/\sqrt{2}$

C $\pm 1/2$

D 0 and 1

$$f(x) = \int_1^x \sqrt{2-t^2}$$

Diff. w.r.t to x

$$\frac{d}{dx} f(x) = \frac{d}{dx} \int_1^x \sqrt{2-t^2}\, dt$$

$$f'(x) = \sqrt{2-x^2} \cdot \frac{d}{dx}(x) - \sqrt{2-1} \cdot \frac{d}{dx}(1)$$

$$f'(x) = \sqrt{2-x^2} \cdot 1$$

$$\boxed{f'(x) = \sqrt{2-x^2}}$$

$$x^2 - f'(x) = 0 \qquad x^2 - \sqrt{2-x^2} = 0$$

$$(x^2)^2 = \left(\sqrt{2-x^2}\right)^2$$

$$x^2 - f'(x) = 0$$

$$f(x) = \int_{\phi(x)}^{\psi(x)} f(t)\, dt$$

Leibnitz THEOREM

$$\frac{d}{dx} f(x) = \frac{d}{dx} \int_{\phi(x)}^{\psi(x)} f(t)\, dt$$

$$\boxed{f'(x) = f[\psi(x)] \frac{d}{dx} \psi(x) - f[\phi(x)] \frac{d}{dx} \phi(x)}$$

$$x^4 = 2 - x^2$$

$$x^4 + x^2 - 2 = 0$$

(biquadratic eqn)

$x^4 + x^2 - 2 = 0$ (biquadratic equⁿ)

↓ quadratic – Know

$x^2 = t$

$= t^2 + t - 2 = 0$

$\Rightarrow t^2 + 2t - t - 2 = 0$

$\Rightarrow t(t+2) - 1(t+2) = 0$

$\Rightarrow \boxed{t = 1 \quad t = -2}$

$t = 1 \quad x^2 = 1 \quad \boxed{x = \pm 1}$

✗ $t^2 = -2$ reject The Root

→ complex No

$\boxed{x = \pm 1}$



Topic : Single Variable Calculus

#Q. The value of the integral $\int_0^1 \sqrt{\frac{1-x}{1+x}}\,dx$ is

(A) $\frac{\pi}{2}+1$

(B) $\frac{\pi}{2}-1$

(C) -1

(D) 1

$$I=\int_0^1 \sqrt{\frac{1-x}{1+x}}\,dx$$

$$\sqrt{\frac{1-x}{1+x}}=\sqrt{\frac{1-\cos 2\theta}{1+\cos 2\theta}}$$

$x=\cos 2\theta$

$$=\sqrt{\frac{1-1+2\sin^2\theta}{1+2\cos^2\theta-1}}$$

$$=\int_0^{\pi/2}\sqrt{\tan^2\theta}\,d\theta$$

Ans.

$$\frac{\pi}{2}-1$$

$$I = \int_0^1 \sqrt{\frac{1-x}{1+x}}\, dx$$

$$\int_0^1 \sqrt{\frac{1-x}{1+x}} \times \sqrt{\frac{1-x}{1-x}}$$

$$= \int_0^1 \frac{(1-x)}{\sqrt{1-x^2}}\, dx$$

$$= \int_0^1 \frac{1-x}{\sqrt{1-x^2}} = \int_0^1 \frac{1}{\sqrt{1-x^2}}\, dx - \int_0^1 \frac{x}{\sqrt{1-x^2}}\, dx$$

$$\int \frac{1}{\sqrt{1-x^2}} = \sin^{-1} x$$

$$\int \frac{x}{\sqrt{1-x^2}}\, dx = \int \frac{-dt}{2\sqrt{t}} = -\frac{1}{2} \times 2\sqrt{t}$$

$$= -\sqrt{1-x^2}$$

$1-x^2 = t$

$x\, dx = -\frac{dt}{2}$

$$\frac{t^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} = \frac{t^{\frac{1}{2}}}{\frac{1}{2}} = 2\sqrt{t}$$

$$= \left[\sin^{-1} x\right]_0^1 - \int_0^0 \frac{x}{\sqrt{1-x^2}}\, dx$$

$$= \left[\sin^{-1} x\right]_0^1 + \int_0^1 \frac{x}{\sqrt{1-x^2}}\, dx = \frac{\pi}{2} - 1$$



Topic : Single Variable Calculus

$$\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$$

#Q. The value of $I = \int_{\pi/4}^{3\pi/4} \frac{\phi}{1+\sin\phi}\,d\phi$ is .....

$$I = \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{(\pi-\phi)}{1+\sin(\pi-\phi)}\,d\phi$$

$$2I = \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{\phi+(\pi-\phi)}{1+\sin\phi}\,d\phi$$

$$2I = \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{\pi}{1+\sin\phi}\,d\phi \qquad I = \frac{\pi}{2}\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{1}{1+\sin\phi}\,d\phi$$

$$\int \frac{1}{1+\sin\phi} \times \frac{1-\sin\phi}{1-\sin\phi} = \int \frac{1-\sin\phi}{1-\sin^2\phi}\, d\phi$$

$$= \int \frac{1-\sin\phi}{\cos^2\phi}\, d\phi = \int \sec^2\phi\, d\phi - \int \sec\phi\tan\phi\, d\phi$$

$$= \underline{\tan\phi - \sec\phi + C}$$

$$I = \frac{\pi}{2}\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{1}{1+\sin\phi}\, d\phi$$

$$= \frac{\pi}{2}\left[\tan\phi - \sec\phi\right]_{\frac{\pi}{4}}^{\frac{3\pi}{4}}$$

$$= \frac{\pi}{2}\left[\left(\tan\left(\frac{3\pi}{4}\right) - \sec\left(\frac{3\pi}{4}\right)\right) - \left(\tan\frac{\pi}{4} - \sec\frac{\pi}{4}\right)\right]$$

$=$ —— Do yourself

Ans



Topic : Single Variable Calculus

#Q. The value of $\int_2^3 \frac{\sqrt{x}}{\sqrt{5-x}+\sqrt{x}}dx$ is .....

$= \frac{3-2}{2} = \frac{1}{2}$

Ans.



Topic : Single Variable Calculus

#Q. The value of $\int_1^{e^{37}} \frac{\pi \sin(\pi \ln x)}{x}\, dx = \ldots\ldots$ 2

$\pi \ln x = t$

$\frac{\pi}{x}\, dx = dt$

$\pi \times \ln 1 = t$

$t = 0$

$\pi \times \ln_e e^{37} = t$

$t = 37\pi$

$= \int_{t=0}^{37\pi} \sin t\, dt$

$= [-\cos t]_0^{37\pi}$

$= -\cos 37\pi + \cos 0$

$= -(-1) + 1$

$= 2$



Topic : Single Variable Calculus

2005 Screening

3 – 5 min

#Q. $\int_{-2}^{0}[x^3+3x^2+3x+3+(x+1)\cos(x+1)]dx$ is equal to

$3 = (1+2)$

$$= \int_{-2}^{0}[(x^3+3x^2+3x+(1+2)+(x+1)\cos(x+1)]$$

$$= \int_{-2}^{0}[(x^3+3x^2+3x+1)+2+(x+1)\cos(x+1)]$$

$$= \int_{-2}^{0}[(x+1)^3+2+(x+1)\cos(x+1)]dx$$

$$= \int_{-1}^{1}(t^3+2+t\cos t)dt$$

$x+1=t$

$dx=dt$

$-2+1=t$

$-1=t$

$0+1=t$

$t=1$

$$= \int_{-1}^{1} t^3 + 2 + t\cos t \, dt$$

$$= \int_{-1}^{1} t^3 \, dt + 2\int_{-1}^{1} dt + \int_{-1}^{1} t\cos t \, dt$$

$$= 2 \times 2 \int_{0}^{1} dt$$

$$= 4 \quad \underline{\text{Ans.}}$$

$f(t) = t^3$

$f(-t) = (-t)^3 = -t^3$

odd

$f(t) = t\cos t$

$f(-t) = -t\cos(-t)$

$= -t\cos t$

odd



Topic : Single Variable Calculus

#Q. Given $y = \int_{1}^{x^2} \cos t \, dt$, then $\frac{dy}{dx} =$ $\cos x^2 \cdot 2x$ Ans.

H.W ✓ gate ✓



Topic : Single Variable Calculus

gate

#Q. If $\varphi(x) = \int_0^{x^2} \sqrt{t}\,dt$ then $\frac{d\varphi}{dx} =$ ____.

A $2x^2$

B $\sqrt{x}$

C 0

D 1



Topic : Single Variable Calculus

#Q. The value of the integral is $I = \int_0^{\pi/4} \cos^2 x \, dx$

gate H.W

A $\frac{\pi}{8} + \frac{1}{4}$

B $\frac{\pi}{8} - \frac{1}{4}$

C $\frac{-\pi}{8} - \frac{1}{4}$

D $\frac{-\pi}{8} + \frac{1}{4}$



Topic : Single Variable Calculus

GATE H.W

#Q. The value of the integral $\int_{-\pi/2}^{\pi/2} (x\cos x)\,dx$ is

A 0

B $\pi - 2$

C π

D $\pi + 2$



Topic : Single Variable Calculus

GATE H.W

#Q. What is the value of the definite integral?

$$\int_{-a}^{a} \frac{\sqrt{x}dx}{\sqrt{x}+\sqrt{a-x}}$$

A 0

B $\frac{a}{2}$

C a

D 2a



Topic : Single Variable Calculus

#Q. The value of the definite integral $\int_1^e \sqrt{x} ln(x) dx$ is

A $\frac{4}{9}\sqrt{e^3} + \frac{2}{9}$

B $\frac{2}{9}\sqrt{e^3} - \frac{4}{9}$

C $\frac{2}{9}\sqrt{e^3} + \frac{4}{9}$

D $\frac{4}{9}\sqrt{e^3} - \frac{2}{9}$

$\ln x = t$

$x = e^t$

GATE

$\int_0^1 e^{t/2} t e^t dt$

$= \int_0^1 e^{\frac{3t}{2}} \cdot t \, dt$

H.W



Topic : Single Variable Calculus

#Q. The value of the integral $\int_0^2 \frac{(x-1)^2 \sin(x-1)}{(x-1)^2+\cos(x-1)}dx$ is

H.W

A 3

B 0

C -1

D -2

GATE

$(x-1)=t$

↓ check

$\int_{-1}^{1} \frac{t^2 \sin t}{t^2+\cos t}dt$

↓

0



Topic : Single Variable Calculus

#Q. $\int_{1/\pi}^{2/\pi} \frac{\cos(1/x)}{x^2} dx = ____.$

H.W

$\frac{1}{x} = t$



Topic : Single Variable Calculus

#Q. Consider the following definite integral $I = \int_0^1 \frac{\left(\sin^{-1} x\right)^2}{\sqrt{1-x^2}} dx$

The value of the integral is

GATE

$\sin^{-1} x = t$

A $\frac{\pi^3}{24}$

B $\frac{\pi^3}{12}$

C $\frac{\pi^3}{48}$

D $\frac{\pi^3}{64}$



Topic : Single Variable Calculus

#Q. The value of $\int_0^{\pi/4} x\cos\left(x^2\right)dx$ correct to three decimal place (assuming that $\pi = 3.14$) is____.

GATE

$x^2 = t$

H.W



Topic : Single Variable Calculus

GATE

#Q. The value of the integral $\int_{-\pi/2}^{\pi/2} \left(x^2 + \ln\frac{\pi + x}{\pi - x}\right)\cos x dx$ is

A 0

B $\frac{\pi^2}{2} - 4$

C $\frac{\pi^2}{2} + 4$

D $\frac{\pi^2}{2}$

sin|cosx

special Integrals

$$\frac{(\text{Power}-1)}{(\text{Power})}$$

1) $I = \int_0^{\frac{\pi}{2}} \cos^7 x\,dx$

$\Rightarrow \frac{6\times4\times2}{7\times5\times3}\times1 = \frac{48}{105}$

$I = \int_0^{\frac{\pi}{2}} \sin^{8}\text{(even)} x\,dx = \frac{7\times5\times3\times1}{8\times6\times4\times2}\times\frac{\pi}{2}$ Ans

$I = \int_0^{\frac{\pi}{2}} \sin^{2}\text{(even)} x\,dx = \frac{1}{2}\times\frac{\pi}{2} \Rightarrow \frac{\pi}{4}$

$I = \int_0^{\frac{\pi}{2}} \cos^5 x\,dx = \frac{4\times2}{5\times3}\times1 = \frac{8}{15}$

$I = \int_0^{\pi/2} \cos^{10} x\,dx = \frac{9\times7\times5\times3\times1}{10\times8\times6\times4\times2}\times\frac{\pi}{2}$

Restrictions

A) Limit always 0 to $\frac{\pi}{2}$

B) In question
even Power = $\frac{\pi}{2}$
odd Power = 1

multiplying factor

IInd type : $I = \int_0^{\frac{\pi}{2}} \sin^m x \cos^n x \, dx$

m —	
n —	
Power (m+n)	

$I = \int_0^{\frac{\pi}{2}} \sin^4 x \cos^6 x \, dx = \frac{3 \times 1 \times 5 \times 3 \times 1}{10 \times 8 \times 6 \times 4 \times 2} \times \frac{\pi}{2}$

m = even
n = even
m+n = even

1) M Even + n even + (m+n even)

Three conditions Working Together = multiply via $\frac{\pi}{2}$

$I = \int_0^{\frac{\pi}{2}} \sin^7 x \cos^8 x \, dx = \frac{6 \times 4 \times 2 \times 7 \times 5 \times 3 \times 1}{15 \times 13 \times 11 \times 9 \times 7 \times 5 \times 3 \times 1} \times 1$

2) other conditions = No multiply

[m even n odd
m odd n odd] No

$I = \int_0^{\frac{\pi}{2}} \sin^3 x \cos^3 x \, dx = \frac{2 \times 2}{6 \times 4 \times 2} \times 1$

$= \frac{1}{12}$



Topic : Single Variable Calculus

#Q. The solution for $\int_0^{\pi/6} \cos^4 3\theta \sin^3 6\theta d\theta$ is

$3\theta = t$

$3 \times 0 = t$

$t = 0$

$3 \times \frac{\pi}{6} = t$

$t = \frac{\pi}{2}$

(A) 0

(B) $\frac{1}{15}$ ✓

(C) 1

(D) $\frac{8}{3}$

$3\theta = t \quad 3 \times 0 = t \quad t = 0 \quad t = \frac{\pi}{2}$

$3\,d\theta = dt \quad d\theta = \frac{dt}{3}$

$$\int_0^{\frac{\pi}{2}} \cos^4 t \sin^3 2t \cdot \frac{dt}{3}$$

$$= \int_0^{\frac{\pi}{2}} \cos^4 t (2 \sin t \cos t)^3 \frac{dt}{3}$$

$$= \frac{8}{3} \int_0^{\frac{\pi}{2}} \sin^3 t \cos^7 t \, dt$$

$$= \frac{8}{3} \times \frac{2 \times 6 \times 4 \times 2}{10 \times 8 \times 6 \times 4 \times 2}$$

$$= \frac{1}{15}$$



Topic : Single Variable Calculus

#Q. The value (round off to one decimal place) if $\int_{-1}^{1} xe^{|x|}dx$ is

GATE

H.W



2 mins Summary

- Topic One
- Topic Two
- Topic Three
- Topic Four
- Topic Five

Single vari — Integral

THANK - YOU